Markov Chain Monte Carlo without Metropolis-Hastings

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Abstract

In a recent Twitter thread, I addressed the issue of whether Metropolis-Hastings corrections are necessary in all cases, when implementing Markov Chain Monte Carlo algorithms. I reproduce this thread here, with additional comments and references.

1 Basics of the Metropolis-Hastings Algorithm

The Metropolis-Hastings algorithm is a Markov Chain Monte Carlo technique for sampling from an unnormalised distribution $\pi(x)$. To achieve this, one uses some *proposal kernel* $q(x \to y)$ to propose moves around the space, and then stochastically accepts and rejects these moves, according to a rule which ensures that one asymptotically samples from the correct invariant measure. One implements such an algorithm by iterating the steps in Algorithm 1.

Algorithm 1 Metropolis_Hastings_Step$(x; \pi, q)$

1. Propose a move to $y \sim q(x \to y)$.
2. Evaluate the *acceptance ratio*, given by
$$\alpha(x \to y) = g\left(\frac{\pi(y)q(y \to x)}{\pi(x)q(x \to y)}\right) \tag{1}$$
where $g(t) = \min(1, t)$.
3. Simulate $u \sim \text{Unif}[0, 1]$, and set $b = \mathbf{I}[u < \alpha]$.
4. If $b = 1$, move to y. Otherwise, remain at x.

The original algorithm (for symmetric proposals q) was proposed in [MRR+53], and then generalised further in works such as [Has70, Gre95, Tie94].

If one uses arbitrary proposals q and **does not** carry out this accept/reject step, i.e. one always moves to the proposed y, then the Markov chain so produced will generally **not** admit π as an invariant measure. As such, the key role of the accept/reject step is to ensure that one converges to the correct invariant measure, and this is generally why they are used.

The contention of this note is to argue that in some cases, even when it is possible to implement these corrections, there might be good reasons to do something else instead. Sometimes, this 'something else' is simply 'don't bother with the Metropolis-Hastings step'. Often, it is beneficial to perform an additional procedure, with a mind towards reducing or controlling the bias of the original Markov chain. In what follows, I will also present instances of some such procedures.

2 When should one correct?

2.1 When corrections are necessary

To begin with: if your proposals are bad, then adjusting (in some form or another) is essential. In particular, if your proposals don't eventually incorporate information about the distribution which you're trying to sample from, you have no hope of sampling successfully. As such, if you're using

something like Random Walk-based, or purely prior-informed proposals (e.g. the pCN algorithm of [BRSV08]), and if you want to converge to something like the right target measure, then you need to involve the likelihood in some way or another. Applying the Metropolis-Hastings correction is a fair way to do this.

2.2 When corrections are negotiable

However, once you start to use more advanced proposals, there are other things working in your favour. If your proposals come from approximating the dynamics of some continuous-time process, such as an ODE [Nea11, Bet17], an SDE [RT96, DM16], or otherwise [BCVD18, BFR19, Şim17, ŞZTG20], then it is typically the case that this (idealised) continuous-time process which would be sampling unbiasedly from π. As such, provided that your numerical approximation of these dynamics is stable, and tracks the true dynamics in a suitable fashion, then your discretised approximation will have *some* invariant measure, which is (hopefully) close to the target which you care about. This can be made precise using techniques like backward error analysis, modified equations, and weak error analysis of SDEs (see e.g. [TT90, Rei99, Zyg11]).

In this regime, one has some options. One natural thing to try is to increase the step-size in your discretisation to be as large as possible (subject to the numerical dynamics remaining stable), employ the discretisation to sample approximately, and simply accept the bias in your approximation. Depending on your model and application, this bias may be acceptable. For example, in certain high-dimensional problems encountered in Bayesian Image Analysis, both theory and practice suggest that this can be the case, see e.g. [Per16, DMP18, VPZ19].

Going further, one can often refine these approximations to account for the induced bias. One relatively transparent way of doing this is with Multi-Level Monte Carlo (MLMC, see [Gil15] for a tutorial review). By coupling the paths of two approximate chains with different step-sizes, it is often possible to efficiently correct for the bias. MLMC was developed by the numerical analysis community, and operates by trading off between the bias, variance, and cost of randomised estimation schemes in a principled way. In constructing an MLMC estimator, one essentially asks: 'given a fixed computational budget, how do I minimise the MSE of my estimator?'. It is perhaps worth highlighting similarities between the bias-variance tradeoff which appears in the analysis of randomised estimators, and the competing desires for numerical integrators to have both high order and large stability regions.

As a side remark, it is worth highlighting that MLMC interfaces quite nicely with parallel computing, as you can effectively distribute the task of correcting for the bias across multiple machines. If this is a resource which is available to you, then MLMC starts to look more attractive.

In any case, the MLMC method is not burdened by a desire to have asymptotically exact samples, which has historically been quite prevalent in the Statistical community. There are good reasons for this (for one thing, it makes designing modular algorithms easier), but it need not be the be-all and end-all.

2.3 Some Examples of 'Acceptable' Inexact Chains

In this subsection, I describe a few situations in which inexact chains can be preferable to their Metropolised alternatives.

2.3.1 Diffusions

If one is using proposals based on Langevin dynamics (either overdamped, as in [RT96], or underdamped, as in [Hor91, DRD18]), then it is reasonable to consider not adjusting, particularly if this allows for larger time steps to be used (see [VPZ19] for work which attacks this problem directly). Applying MLMC to proposals based on SDE dynamics is also relatively transparent, and has been understood since at least the work of [Gil08].

2.3.2 Hamiltonian Dynamics

If one is using proposals based on Hamiltonian dynamics, as in Hamiltonian Monte Carlo (HMC) [Nea11, Bet17], then the appeal of inexact methods depends on the precise instance of HMC which you are using. For 'vanilla' HMC, not Metropolising might allow for the use of an increased step-size, which could improve mixing.

By contrast, when using one of the trajectorial variants of HMC (e.g. NUTS, [HG14]), then this is not really the case. In particular, trajectorial versions of HMC operate by i) generating a trajectory through the current point, and ii) selecting a new point from that trajectory. The selection procedure involves weights which are related to the Metropolis-Hastings correction, and so it is harder to avoid them. Essentially, if you are using a trajectorial method, then increasing the step-size too much can cause the selection step to degenerate, thus negating the benefits of allowing oneself to be inexact.

2.3.3 Large-Data Problems

If one is interested in doing MCMC on posterior distributions involving large datasets, it is by now common to consider using proposals which rely upon data subsampling (e.g. [WT11, CFG14]). In this case, there is a strong argument for avoiding Metropolis-Hastings corrections, due to their cost. Moreover, one can correct for the bias of using uncorrected dynamics using the techniques of [GNS$^+$16, DSM$^+$16] and variations thereof. This can be particularly appealing when using smarter numerical integration schemes to implement the dynamics, e.g. [SZLS15, LS16, MW18]. In some sense, the 'big data' setting is a great fit for Multi-Level methods. The cost of doing Metropolis-Hastings corrections gets worse and worse as the size of the dataset grows, but the difficulty of approximating the dynamics of an SDE remains comparatively stable.

2.3.4 Gibbs Sampling

A slightly different example is the Gibbs sampler [GG84]. These differ in that they tend to be designed to provide asymptotically-exact samples right from the start, with no need for a Metropolis-Hastings correction. Knowing this, it is perhaps not clear what role biased methods might have to play. The key is that one can implement Gibbs samplers *incorrectly*, in ways which might be faster (in real time), and that this may be worthwhile. In particular, instead of updating the variables on a site-by-site basis, one can instead implement parallel updates, i.e. resample $x_i|\{x_j\}_{j\neq i}$ for all i simultaneously (see work on *probabilistic cellular automata*, e.g. [DPSS12, DPSS15, LS13, PSS16]). This incurs a bias, but can speed up mixing, particularly if the updates can be distributed appropriately. Note that in this case, one still needs to be careful vis-a-vis the stability of the resulting Markov chain, which can fail to be ergodic without additional care [JSW13]. In this setting, one can again make use of MLMC methods. One route to this is to first recast Gibbs sampling as a continuous-time Markov process, and then simulate this process approximately using tau-leaping [Gil01, MH19]. The Multi-Level procedure of [AH12] can then be applied.

3 Conclusion

Beyond MLMC, there are of course various other approaches which one can use to correct for bias, such as post-processing methods for SDEs [Vil15], post-hoc sample reweighting [LL16, HSR20], and others. Regretfully, I cannot survey all of these approaches, though I comment briefly that work in numerical analysis tends to be very pragmatic about such matters.

In closing, if one is using the Metropolis-Hastings algorithm, and the proposals are uninformed, then it is clear that some sort of adjustment needs to take place eventually. However, if the uncorrected proposals would already mix rapidly, to some distribution which is close to the distribution you would like, then there might be better ways of addressing the inexactness than using Metropolis-Hastings corrections.

As a final, more general remark, it bears mentioning that one secret of Multi-Level Monte Carlo (and its variants, such as Multi-Index Monte Carlo [HANT16]) is that even if one has access to an *exact* estimator of some quantity of interest, it is often worth one's while to find a *worse* estimator which is cheaper to implement, and then re-allocate computational resources accordingly. Paying full price for an exact solution is not always the sure-thing victory which it might appear to be.

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